STEP 1
Order Entry
ERP sales order

▲ Queue: awaiting credit/release

DECISION

Can we commit to the requested date?

QUESTIONS

Was the order placed before the daily cutoff?

Is the requested date feasible?

VARIABLES

- ✓ Order time of day S1 T
  `order date`
  ↳ *ERP order header*
- ✓ Payment type S1
  `Type`
  ↳ *ERP order header*
- ✓ Price, discount, quantity S1
  `Sales…`
  ↳ *ERP order lines*
- ✓ Customer segment, market S1
  `Market`
  ↳ *CRM customer master*
- ◐ Requested vs. promised date gap S1 T
  ↳ *ERP order header*
- ◐ Available-to-promise qty S1
  ↳ *ERP inventory and ATP*

Primary system: ERP sales order

STEP 2
Credit & Release
ERP order status

▲ Queue: awaiting wave/pick

DECISION

Release the order now or hold it?

QUESTIONS

How complex is the order?

How long has it waited for release?

VARIABLES

- ✓ Lines in order (derived) S1
  `Order Id`
  ↳ *ERP order lines*
- ◐ Order-to-release hours S2 T
  ↳ *ERP status log*
- ◐ Slack days to promise date S2 T
  ↳ *ERP order header*
- ! Order status at extract After
  `Order Status`
  ↳ *ERP status at extract time*

Primary system: ERP status change log

STEP 3
Pick & Pack
WMS

▲ Queue: awaiting tender

DECISION

Prioritize this order in the wave?

QUESTIONS

Is the network running behind?

Which facility ships it?

VARIABLES

- ✓ Regional order load, prior 7 days (derived) S1 T
  `order date`
  ↳ *Derived from ERP order history*
- ◐ Pick backlog at release S3 T
  ↳ *WMS task queue*
- ◐ Pick and pack timestamps S3 T
  ↳ *WMS scan events*
- ✗ Shipping facility S3
  ↳ *Not recorded*
- ✗ Shipment weight S3
  ↳ *WMS pack station*

Primary system: WMS task records

STEP 4
Carrier Tender
TMS

▲ Queue: awaiting pickup

DECISION

Expedite, switch carrier, or warn the customer?

QUESTIONS

Is the service level achievable?

Is this lane running late?

VARIABLES

- ✓ Shipping mode S1
  `Shipping Mode`
  ↳ *ERP order header*
- ✓ Scheduled shipping days S1
  `Days for shipment (scheduled)`
  ↳ *ERP order header*
- ✓ Lane: mode x region (derived) S1
  `Order Region`
  ↳ *Derived from ERP order and region*
- ◐ Carrier and tender acceptance S1
  ↳ *TMS tender history*
- ◐ Hours from tender to pickup S4 T
  ↳ *TMS appointments*

Primary system: TMS tender records

STEP 5
Pickup & Transit
Carrier EDI 214 / visibility

▲ In transit

Limited options

DECISION

Intervene while in transit?

QUESTIONS

Did it ship on time?

Is the ETA slipping?

VARIABLES

- ! Actual ship date After
  `shipping date`
  ↳ *WMS ship confirmation*
- ! Actual shipping days After
  `Days for shipping (real)`
  ↳ *WMS ship confirmation*
- ✗ Milestone ETA variance S5 T
  ↳ *Carrier portal or EDI 214*
- ✗ Weather and congestion S5
  ↳ *External data feed*

Primary system: carrier milestones

OUTCOME (Y)
Delivered On Time?
Proof of delivery

Too late to act

TARGET

Shipment late (1) or not (0)

VARIABLE

- ✓ Target
  `Late_delivery_risk`
  ↳ *Proof of delivery*

Known only after the outcome. Exclude.

- ! Label restated After
  `Delivery Status`
  ↳ *Derived from proof of delivery*
- ! Actual days define the label

Population rule. 7,754 canceled shipments have no delivery outcome and are removed.

Identifiers and PII excluded: names, email, password, street, IDs

Decision window: risk score must be available here to expedite, re-sequence, or re-tender

✓ In the dataset (or derived from it) ◐ Captured elsewhere in the organization, missing from dataset

✗ Not captured ! Timing violation (post-outcome) ↳ *Where the data stream lives*

S1 Step where the value is created T Timing and queue variable ▲ Wait / queue

9 of 21
predictor variables available

4
post-outcome fields flagged

2 of 9
timing and queue variables